

Unit 3 Question Packet
Atoms, Ions, & Isotopes

Name**KEY**.....
Period

SKILLS

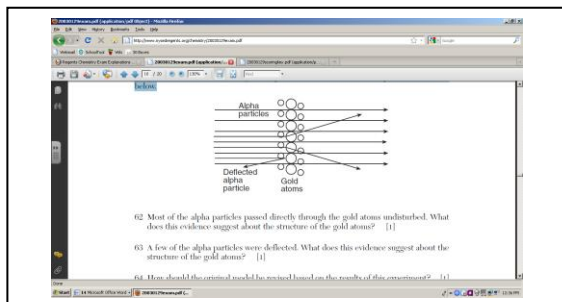
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| 1. interpreting MODELS OF THE ATOM | 4. interpreting ISOTOPIC NOTATION |
| 2. determining # of SUBATOMIC PARTICLES
IN ATOMS | 5. CALCULATING ATOMIC MASS |
| 3. identifying GROUND-STATE & EXCITED-STATE
ELECTRON CONFIGURATIONS | 6. determining # of SUBATOMIC PARTICLES
IN IONS |

SKILL #1: interpreting MODELS OF THE ATOM

- refer to your notes and RB p. 33-34

1. One model of the atom states that atoms are tiny particles composed of a uniform mixture of positive and negative charges. Scientists conducted an experiment where alpha particles were aimed at a thin layer of gold atoms.

Most of the alpha particles passed directly through the gold atoms. A few alpha particles were deflected from their straight-line paths. An illustration of the experiment is shown below.



a. Most of the alpha particles passed directly through the gold atoms undisturbed. What does this evidence suggest about the structure of the gold atoms?

Mostly empty space

b. A few of the alpha particles were deflected. What does this evidence suggest about the structure of the gold atoms?

Small, dense, positively-charged center (nucleus)

c. How should the original model be revised based on the results of this experiment?

The atom has a positively charged nucleus; negative electrons surround the outside. The positive charges are in the nucleus; electrons are not mixed in the nucleus. nucleus smaller than atom

2. In 1897, J. J. Thomson demonstrated in an experiment that cathode rays were deflected by an electric field. This suggested that cathode rays were composed of negatively charged particles found in all atoms. Thomson concluded that the atom was a positively charged sphere of almost uniform density in which negatively charged particles were embedded. The total negative charge in the atom was balanced by the positive charge, making the atom electrically neutral.

In the early 1900s, Ernest Rutherford bombarded a very thin sheet of gold foil with alpha particles. After interpreting the results of the gold foil experiment, Rutherford proposed a more sophisticated model of the atom.

a. State one conclusion from Rutherford's experiment that contradicts one conclusion made by Thomson.

An atom has a nucleus that is positively charged; An atom is mostly empty space. Negatively charged particles are located outside the positive nucleus.

b. State one aspect of the modern model of the atom that agrees with a conclusion made by Thomson.

An atom has equal amounts of negative and positive charge.

An atom has an equal number of protons and electrons.

All atoms contain electrons.

Electrons are negatively charged.

SKILL #2: determining # of SUBATOMIC PARTICLES
and the Periodic Table

- refer to your notes, RB p. 35-37, Table S,

3. Complete the table below based on information provided for each atom.

Remember: all atoms are electrically neutral.

Atom	Protons	Electrons	Neutrons	Mass #	Atomic #	Nuclear charge	# of Nucleons	Nuclear diagram	Element's Symbol
A	44	44	58	102	44	+44	102	58 n 44 p	Ru
B	84	84	125	209	84	+84	209	125 n 84 p	Po
C	25	25	31	56	25	+25	56	31 n 25 p	Mn
D	89	89	140	229	89	+89	229	140 n 89 p	Ac
E	28	28	30	58	28	+28	58	30 n 28 p	Ni
F	92	92	141	233	92	+92	233	141 n 92 p	U
G	56	56	82	138	56	+56	138	82 n 56 p	Ba

SKILL #4: Interpreting ISOTOPIC NOTATION

- refer to your notes and RB p. 38

8. Copper has two naturally-occurring isotopes, Cu-63 and Cu-65. State, in terms of subatomic particles, how an atom of Cu-63 differs from an atom of Cu-65.

An atom of copper-63 has two fewer neutrons than an atom of copper-65. Or...

An atom of Cu-63 has 34 neutrons and an atom of Cu-65 has 36 neutrons.

SKILL #5: CALCULATING ATOMIC MASS

- refer to your notes and RB p. 39

10. Calculate the atomic mass for each element below, given the relative abundances and the mass numbers of the isotopes. Show ALL work. (GSSC)

a. 19.78% of B-10
80.22% of B-11

G
S: decimal x mass = products (add)
S: $0.1978 \times 10 = 1.978$
 $0.8022 \times 11 = 8.8242$
10.80 amu
C: 10.80 is between 10 and 11, closer to 11 (80%)

b. 78.70% of Mg-24
10.13% of Mg-25
11.17% of Mg-26

G
S: decimal x mass = products (add)
S: $0.7870 \times 24 = 18.888$
 $0.1013 \times 25 = 2.5325$
 $0.1117 \times 26 = 2.9042$
24.32 amu
C: 24.32 is between 24 and 26, closer to 24 (79%)

c. 93.12% of K-39
6.88% of K-41

G
S: decimal x mass = products (add)
S: $0.9312 \times 39 = 36.3168$
 $0.0688 \times 41 = 2.8208$
39.1 amu
C: 39.1 is between 39 and 41, closer to 39 (93%)

11. Base your answers to the following questions on the data table below, which shows three isotopes of neon.

Isotope	Atomic Mass (atomic mass units)	Percent Natural Abundance
^{20}Ne	19.99	90.9%
^{21}Ne	20.99	0.3%
^{22}Ne	21.99	8.8%

Of atomic particles, state one difference between these three isotopes of neon. [1]
the atomic masses and the natural abundances shown in the data table, in the
vided in your answer booklet, show a correct numerical setup for calculating

a. In terms of atomic particles, state one difference between these three isotopes of neon.
different number of neutrons

b. Based on the atomic masses and the natural abundances shown in the data table, show a correct numerical setup for calculating the average atomic mass of neon.

S: decimal x mass = products (add)
S: $0.909 \times 19.99 = 18.17091$
 $0.003 \times 20.99 = 0.06297$
 $0.088 \times 21.99 = 1.93512$
20.17 amu
C: 20.17 is between 19.99 and 21.99, closer to 19.99 (91%)

c. Based on natural abundances, the average atomic mass of neon is closest to which whole number? **20**

12. Naturally occurring elemental carbon is a mixture of isotopes. The percent composition of the two most abundant isotopes is listed below.

- **98.93%** of the carbon atoms have a mass of **12.00** atomic mass units.
- **1.07%** of the carbon atoms have a mass of **13.00** atomic mass units.

a. Show a correct **numerical setup** for calculating the average atomic mass of carbon.

S: **decimal x mass = products (add)**

S: **0.9893 x 12.00 = 11.8716**

0.0107 x 13.00 = 0.1391

12.01 amu

C: 12.01 is between 12 and 13, closer to 12 (99%)

b. Describe, in terms of *subatomic particles found in the nucleus*, one difference between the nuclei of carbon-12 atoms and the nuclei of carbon-13 atoms. The response must include both isotopes.

Carbon-13 has one more neutron than carbon-12.

13. The atomic mass of element A is 63.6 atomic mass units. The only naturally occurring isotopes of element A are A-63 and A-65. The percent abundances in a naturally occurring sample of element A are closest to

(1) 31% A-63 and 69% A-65

(2) 50% A-63 and 50% A-65

(3) 69% A-63 and 31% A-65

(4) 100% A-63 and 0% A-65

14. A 100.00-gram sample of naturally occurring boron contains 19.78 grams of boron-10 (atomic mass = 10.01 atomic mass units) and 80.22 grams of boron-11 (atomic mass = 11.01 atomic mass units). Which numerical setup can be used to determine the atomic mass of naturally occurring boron? **1**

SKILL #6: **Determining # of SUBATOMIC PARTICLES
IN IONS**

– refer to your notes, RB p. 44-45, & P.T.

15. Complete the table below.

Ion	Charge	# of protons	# of electrons	Electron configuration	E.C. same as...	Ion larger or smaller than atom?
Al ³⁺	+3	13	10	2-8	Ne	smaller
P ³⁻	-3	15	18	2-8-8	Ar	larger
Sr ²⁺	+2	38	36	2-8-18-8	Kr	smaller
Na ⁺	+1	11	10	2-8	Ne	smaller

Mg ²⁺	+2	12	10	2-8	Ne	smaller
H ⁺	+1	1	0	0	a proton	smaller
O ²⁻	-2	8	10	2-8	Ne	larger
Cl ⁻	-1	17	18	2-8-8	Ar	larger
K ⁺	+1	19	18	2-8-8	Ar	smaller
F ⁻	-1	9	10	2-8	Ne	larger

16. Explain, in terms of subatomic particles, why an oxygen atom is electrically neutral.
 An atom of oxygen has 8 protons and 8 electrons.

17. Explain, in terms of subatomic particles, why an oxide ion, O²⁻, has a negative charge.
 An oxide ion has 2 more electrons than protons, giving it a charge of -2.

18. Compare the number of protons to the number of electrons in a positive ion.
 A positive ion has more protons than electrons

19. Compare the number of protons to the number of electrons in a negative ion.
 A negative ion has more electrons than protons

20. Explain, in terms of subatomic particles, why a chlorine ion is larger than a chlorine atom.
 A chlorine ion has one more electron than a chlorine atom.